

CASEFILE: An AI-Powered Missing Person Investigation and Probable Location Prediction System Using Advanced Machine Learning

1. Abstract

Missing person cases are a major challenge for law enforcement agencies because locating a missing individual often requires analyzing large amounts of information from different sources. Traditional investigation methods mainly depend on manual searching, witness statements, records, and investigation experience, which can be time-consuming and difficult to scale.

CASEFILE is an AI-powered missing person investigation and probable location prediction system developed using Advanced Machine Learning techniques. The system analyzes important information related to a missing person, such as age, gender, last known location, date of disappearance, behavioral patterns, physical characteristics, and other case-related attributes. Machine Learning models are used to identify patterns in historical cases and generate a probable location or location category where the missing person may potentially be found.

The system also provides investigation-oriented insights and probability-based predictions that can assist investigators in prioritizing search areas. Explainable Machine Learning techniques can be used to understand which features have contributed most to a prediction. The objective is not to replace investigators but to provide an intelligent decision-support system that can make investigations faster, more systematic, and data-driven.

2. Introduction

Missing person investigations are complex because every case can involve different circumstances, locations, personal characteristics, and behavioral patterns. Investigators may need to examine previous cases, geographical information, demographic details, timelines, and other evidence before deciding where to focus their search.

With the development of Artificial Intelligence and Machine Learning, historical investigation data can be analyzed to discover patterns that may not be easily identified through manual analysis.

CASEFILE is proposed as an intelligent investigation support system that combines missing-person case data with Machine Learning algorithms. The system processes historical and current case information and attempts to predict a probable location or location category for a missing person.

The system can be useful for:

- Analyzing historical missing-person cases
- Identifying patterns in missing-person data
- Predicting probable search locations
- Prioritizing cases based on risk or probability
- Supporting investigators in decision-making
- Providing interpretable reasons behind predictions
- Reducing the time required for initial case analysis

The system is designed as a **decision-support tool**, meaning that its predictions should support human investigation rather than replace police officers, investigators, or other authorized authorities.

3. Problem Statement

Traditional missing-person investigations can require considerable time because investigators have to manually analyze multiple attributes and compare the current case with historical records.

The major problems include:

1. Large amounts of case data are difficult to analyze manually.
2. Historical patterns may remain hidden in large datasets.
3. Search areas may be very large.
4. Investigators may have limited time and resources.
5. Similar previous cases may be difficult to identify quickly.
6. Manual analysis can result in inconsistent prioritization of search areas.
7. There is a need for an intelligent system that can provide data-driven predictions.

Therefore, the problem addressed by CASEFILE is:

To develop an AI-powered Machine Learning system capable of analyzing missing person case information and predicting a probable location or location category to assist investigators in prioritizing search efforts.

4. Objectives

The major objectives of the CASEFILE project are:

Primary Objectives

- To develop an AI-powered missing-person investigation system.
- To analyze historical missing-person case records.
- To preprocess and transform investigation data for Machine Learning.
- To identify important factors affecting probable location.
- To train Machine Learning models for prediction.
- To compare different ML algorithms.
- To generate probable location predictions.
- To provide interpretable and explainable predictions.

Secondary Objectives

- To reduce the time required for preliminary case analysis.
- To assist investigators in prioritizing search locations.
- To identify similarities between new and historical cases.
- To provide a user-friendly investigation interface.
- To improve the systematic use of historical investigation data.

5. Literature Review

Artificial Intelligence has increasingly been applied to crime analysis, geographical prediction, missingperson investigations, and law-enforcement decision support.

Traditional approaches to missing-person investigations generally rely on manual searches, geographical analysis, witness information, public announcements, and

investigation databases. Although these methods are important, they may become less efficient when the amount of available information increases.

Machine Learning provides an opportunity to analyze historical records and identify relationships between different attributes. Classification algorithms can categorize cases into predefined location classes, while regression techniques can estimate numerical outcomes. Clustering techniques can also identify groups of similar cases without requiring predefined labels.

Several Machine Learning approaches commonly used in predictive analytics include:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- K-Nearest Neighbors
- Gradient Boosting
- XGBoost
- Clustering algorithms

Among these techniques, tree-based ensemble methods such as Random Forest and Gradient Boosting are particularly useful when datasets contain a combination of numerical and categorical features.

Explainable AI is also important in sensitive domains such as law enforcement. Investigators should be able to understand why a model produced a particular prediction rather than relying entirely on a black-box result.

CASEFILE combines these concepts into an investigation-oriented framework for missing-person analysis and probable location prediction.

6. Dataset Description

The dataset used by CASEFILE contains historical missing-person case information. Each record represents a missing-person case and contains different attributes that can be useful for identifying patterns and predicting probable locations.

Typical attributes considered by the system include:

Feature	Description
Case Number	Unique identifier of the case
Age	Age of the missing person
Gender	Gender of the person
Last Known Location	Location where the person was last observed
Date Missing	Date on which the person was reported missing
Time Information	Relevant time-related information
Physical Characteristics	Available physical information
Case Characteristics	Information related to the case
Status	Current case status, where available

Location Category Target variable representing probable
location/category

The dataset may contain both **numerical and categorical variables**. Numerical variables can include age and time-related measurements, while categorical variables can include gender, region, case type, and other attributes.

For Machine Learning, the dataset is divided into:

- **Input features (X):** Information about the missing-person case.
- **Target variable (Y):** Probable location/location category.

The dataset is cleaned and transformed before being provided to the Machine Learning models.

7. Data Preprocessing

Data preprocessing is an important step because real-world investigation data may contain missing values, inconsistent formats, duplicate records, and categorical variables.

The following preprocessing steps are applied:

7.1 Data Cleaning

Duplicate records are identified and removed wherever appropriate.

7.2 Missing Value Handling

Missing values are analyzed before model training.

Depending on the feature:

- Numerical missing values can be replaced using mean or median.
- Categorical missing values can be replaced using mode or an "Unknown" category.
- Records with critical missing target values may be excluded.

7.3 Data Type Conversion

Dates, ages, numerical values, and categorical values are converted into appropriate data types.

7.4 Categorical Encoding

Machine Learning algorithms generally require numerical input. Therefore, categorical attributes are converted into numerical representations using techniques such as:

- Label Encoding
- One-Hot Encoding

7.5 Feature Scaling

Algorithms such as KNN and SVM can benefit from feature scaling. Numerical features can therefore be standardized using StandardScaler or another appropriate scaling method.

7.6 Train-Test Split

The processed dataset is divided into training and testing sets.

A commonly used division is:

- **80% Training Data**
- **20% Testing Data**

The training data is used to learn patterns, while the testing data is used to evaluate how well the model performs on unseen cases.

8. Feature Engineering

Feature engineering involves creating useful features from the original case information.

Important engineered features can include:

Age Group

Instead of using only exact age, individuals can be grouped into categories such as:

- Child
- Teenager
- Young Adult
- Adult
- Senior

Time-Based Features

The date of disappearance can be transformed into:

- Day
- Month
- Year
- Day of week

- Season

These features may help identify temporal patterns.

Location Features

The last known location can be transformed into:

- City
- District
- State
- Urban/Rural category
- Geographic region

Case Duration

If recovery information is available, the number of days between disappearance and recovery can be calculated.

Distance-Based Features

If geographic coordinates are available, distances between relevant locations can be calculated.

Feature engineering helps the Machine Learning model understand the dataset more effectively and discover relationships between case characteristics and probable locations.

9. Methodology

The CASEFILE methodology consists of multiple stages.

Step 1: Data Collection

Historical missing-person case records are collected from the available dataset.

Step 2: Data Understanding

The dataset is examined to understand:

- Number of records
- Features
- Data types
- Missing values
- Duplicate values
- Target variable

Step 3: Data Preprocessing

Missing values, duplicates, inconsistent values, and categorical variables are handled.

Step 4: Feature Engineering

Relevant features are selected and new meaningful features are generated.

Step 5: Dataset Splitting

The processed dataset is divided into training and testing datasets.

Step 6: Model Development

Multiple Machine Learning algorithms are trained.

Step 7: Model Evaluation

The models are evaluated using suitable performance metrics.

Step 8: Best Model Selection

The best-performing model is selected based on evaluation results.

Step 9: Prediction

The trained model receives information about a new missing-person case and produces a probable location/category.

Step 10: Explainability

The system provides information about the important features contributing to the prediction.

10. ML Algorithms

Multiple Machine Learning algorithms can be considered for the CASEFILE prediction task.

10.1 Logistic Regression

Logistic Regression is a classification algorithm used to predict the probability of different classes.

In CASEFILE, it can be used as a baseline classification model for predicting location categories.

Advantages

- Simple and easy to interpret
- Fast to train
- Useful as a baseline model

10.2 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees.

Each tree produces a prediction and the final result is obtained by combining the predictions.

Advantages

- Good predictive performance
- Handles complex relationships
- Less prone to overfitting than a single decision tree
- Provides feature importance

Random Forest is particularly suitable for CASEFILE because investigation data may contain many interacting features.

10.3 Support Vector Machine

SVM attempts to find an optimal boundary between different classes.

It can be useful when the data contains complex classification boundaries.

Advantages

- Effective in high-dimensional datasets
- Can use different kernels
- Useful for classification tasks

10.4 K-Nearest Neighbors

KNN predicts a new case based on the most similar cases in the historical dataset.

For CASEFILE, this means a new missing-person case can be compared with historically similar cases.

Advantages

- Simple concept
- Useful for similarity-based analysis
- No complex training process

10.5 Ensemble/Boosting Models

Advanced boosting algorithms such as Gradient Boosting or XGBoost can also be used.

These models build multiple weak learners sequentially and improve prediction performance by focusing on previous errors.

11. Model Training

After preprocessing and feature engineering, the dataset is divided into training and testing sets.

The training dataset is provided to the Machine Learning algorithms.

The general training process is:

1. Load the processed dataset.
2. Separate input features and target variable.

3. Encode categorical features.
4. Scale numerical features where required.
5. Split the dataset into training and testing sets.
6. Initialize the ML model.
7. Train the model using training data.
8. Generate predictions for testing data.
9. Calculate evaluation metrics.
10. Compare the models.

12. Model Evaluation

Model evaluation determines how effectively the trained model performs on unseen data.

For classification-based prediction, the following metrics can be used.

Accuracy

Accuracy measures the percentage of correct predictions.

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions}$$

Precision

Precision measures how many predicted positive cases are actually correct.

Recall

Recall measures how many actual relevant cases are correctly identified.

F1-Score

F1-score combines Precision and Recall.

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Confusion Matrix

A confusion matrix shows the relationship between:

- True Positive
- True Negative
- False Positive
- False Negative

For multi-class location prediction, the confusion matrix can show which location categories are being predicted correctly or incorrectly.

Cross-Validation

K-Fold Cross-Validation can also be used to determine whether the model performs consistently across different subsets of the dataset.

13. Results

The CASEFILE system is designed to compare multiple Machine Learning models and select the most suitable model for probable location prediction.

The expected output of the CASEFILE system is not simply a single location name. Instead, the system can provide a **probable location/category with a confidence or probability score**, allowing investigators to prioritize search areas.

For example:

Predicted Probable Location: Region/Location A

Prediction Probability: XX%

Risk/Priority Level: High

Important Factors: Last known location, age group, time since disappearance, and historical case similarity.

The result should be interpreted as an AI-generated probability rather than a confirmed location.

14. Explainability

Explainability is especially important because CASEFILE is designed for an investigation-related application.

A Machine Learning prediction should not be treated as a final decision without understanding the factors behind it.

The system can use techniques such as:

- Feature Importance
- SHAP
- LIME
- Decision Tree Rules

Feature Importance

Feature importance identifies which variables contributed most to the model's prediction.

For example, important features may include:

- Last known location

- Age
- Gender
- Time since disappearance
- Geographic region
- Case characteristics

15. Limitations

Although CASEFILE can provide useful analytical support, it has several limitations.

1. Machine Learning predictions depend heavily on the quality of the dataset.
2. Historical patterns may not always apply to new cases.
3. Missing or incorrect information can reduce prediction accuracy.
4. Location prediction cannot guarantee the actual location of a missing person.
5. Rare or unusual cases may be difficult for the model to predict.
6. Geographic and behavioral patterns may change over time.
7. Dataset bias can result in biased predictions.
8. A prediction should never be considered proof or evidence by itself.
9. The system cannot replace professional investigation.
10. Real-world cases may contain information that is not available in the dataset.

Therefore, CASEFILE should be treated as an **investigation-support system rather than an autonomous investigation system.**

16. Ethical Considerations

Because CASEFILE deals with sensitive missing-person information, ethical considerations are extremely important. **Privacy**

Personal information should be protected and only authorized users should be allowed to access case data.

Data Security

Sensitive information should be stored securely using appropriate authentication and access-control mechanisms.

Human Oversight

AI predictions should always be reviewed by authorized investigators.

Avoiding Bias

The dataset should be examined for demographic, geographic, or other biases.

Responsible Prediction

The system should clearly communicate that predictions are probabilistic and not confirmed facts.

No Automatic Decisions

The system should not automatically accuse individuals, identify suspects, or make legal decisions based solely on Machine Learning predictions.

Transparency

Users should be able to understand the major factors contributing to a prediction.

Appropriate Use

The system should only be used for legitimate investigation and missing-person assistance purposes by authorized personnel.

17. Future Scope

CASEFILE can be further improved by integrating additional technologies and data sources.

17.1 Real-Time Location Intelligence

The system could integrate authorized real-time geographic information to improve search-area prioritization.

17.2 GIS and Interactive Maps

GIS integration could display:

- Last known location
- Probable search zones
- Historical case clusters
- Geographic risk areas on an interactive map.

17.3 Computer Vision

Computer Vision could potentially assist authorized investigators in comparing available images or identifying visually similar information, subject to strict privacy and legal safeguards.

17.4 NLP-Based Investigation

Natural Language Processing could extract useful information from:

- Investigation notes
- Witness statements
- Case descriptions
- Reports and convert unstructured text into structured features.

17.5 Advanced Deep Learning

Deep Learning models could be explored when sufficiently large and representative datasets become available.

17.6 Temporal Prediction

Time-series models could analyze how probable locations change as the duration of a case increases.

17.7 Similar Case Retrieval

A future version could automatically retrieve historically similar missing-person cases and display their outcomes.

17.8 Mobile Application

A secure mobile application could allow authorized investigators to access predictions and case information in the field.

17.9 Multimodal AI

Future versions could combine:

Text + Location + Images + Historical Case Data + Time

Information to create a more comprehensive

investigation-support system.

18. Conclusion

CASEFILE demonstrates how Artificial Intelligence and Advanced Machine Learning can be applied to support missing-person investigations. The system uses historical case information, data preprocessing, feature engineering, Machine Learning algorithms, model evaluation, and explainability techniques to generate probable location predictions.

The major advantage of the proposed system is its ability to analyze large amounts of historical information and identify patterns that may assist investigators in prioritizing search areas. Algorithms such as Decision Tree, Random Forest, Logistic Regression, KNN, SVM, and boosting methods can be evaluated to identify the most suitable predictive approach.

However, the system must be used responsibly. A Machine Learning prediction represents a probability and should never be considered a confirmed fact. Human investigators, legal procedures, physical evidence, and verified information must remain the primary basis for investigation decisions.

Overall, **CASEFILE provides a foundation for an intelligent, explainable, and data-driven missingperson investigation support system.** With better datasets, GIS integration, NLP, Computer Vision, realtime authorized data, and advanced

Machine Learning techniques, the system can be expanded into a more comprehensive investigation-assistance platform in the future.